UNIVERSITI TEKNOLOGI MARA
MARKING SCHEME

COURSE	:	DATABASE DESIGN AND DEVELOPMENT
COURSE CODE	:	ICT450
EXAMINATION	:	NOVEMBER 2026 (MOCK)
TIME	:	3 HOURS

MARKING SCHEME NOTES

1. This marking scheme accompanies the ICT450 mock paper (November 2026). Answers are unofficial model answers, not official UTM marking schemes.

2. Marking points state the expected answer and the allocation of marks; every question retains its source paper and page reference.

3. Five questions x 20 marks; total = 100 marks.

4. Real-bank questions: 5. Placeholders: 0. Questions and answers are for local study use only.

5. Structure derived from 12 unique ICT450 past papers; every real question remains traceable to its source page.

FOR MARKERS' USE ONLY

This marking scheme consists of 8 printed pages.

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MARKING GUIDE

QUESTION 1 (20 marks)

Source: ICT450-2024-JAN · page [2] · bank id ICT450-2024-JAN-Q1

Reference answer

(a) Meaning of data and database. [4 marks]

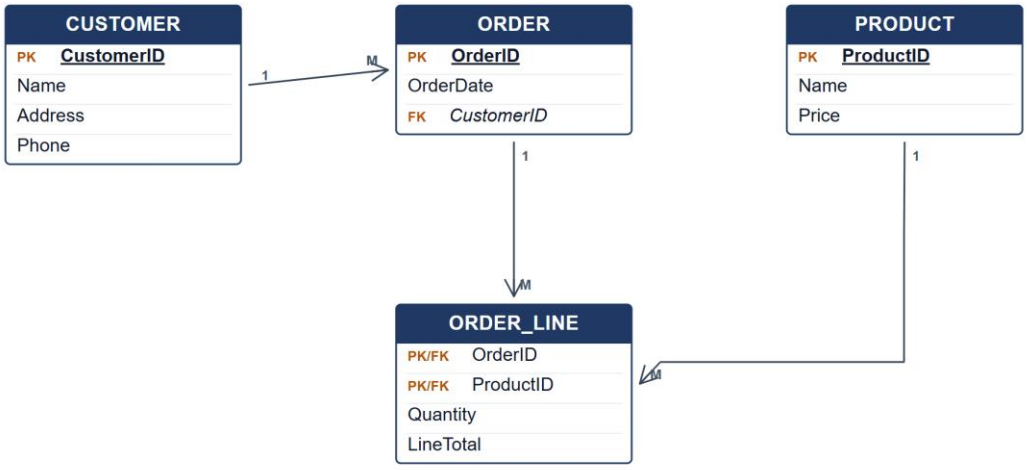
- Data are raw facts, such as a customer name or an order amount, that have not yet been processed into meaning.
- A database is an organised collection of logically related data stored and managed as a single repository.

(b) Data inconsistency and an example. [4 marks]

- Data inconsistency exists when the same fact has different values in different places.
- Example: a customer's phone number is updated in the Customers table but the old number remains in an Orders table.

(c) Zinnirah's problem and the ERD solution. [6 marks]

- The three tables are not linked by keys, so joins become complex and the same customer/product data is duplicated.
- The solution is an ERD with CUSTOMER, ORDER, PRODUCT and ORDER_LINE, where ORDER_LINE resolves the many-to-many order-product relationship.



ERD for the ZY Stationary database.

(d) Three types of SDLC maintenance. [6 marks]

- Corrective maintenance — fixing errors found after delivery.
- Adaptive maintenance — changing the system for a new environment, platform or business rule.
- Perfective maintenance — improving performance or adding enhancements requested by users.

Marking guide

- a) 2 marks each for data and database (4).
- b) 2 marks for the definition and 2 for the example (4).
- c) 2 marks for the problem and 4 marks for the ERD (6).
- d) 2 marks for each maintenance type (6).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the

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ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 1 DATABASE CONCEPT (1).pdf; TOPIC 2 DATA MODEL (1).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- The three tables are Customers, Orders and Products; an order can contain many products.

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QUESTION 2 (20 marks)

Source: ICT450-2023-FEB · page [3] · bank id ICT450-2023-FEB-Q2

Reference answer

(a) Four activities in the database initial study. [8 marks]

- Analyse the company situation — study the organisation's structure, mission, operations, data handling and business environment.
- Define the problems and constraints — identify weaknesses, missing or redundant data, security issues and limits such as budget, time or existing systems.
- Define the objectives — state what the database must achieve and check that the objectives match the end users' needs.
- Define the scope and boundaries — decide which business areas, entities, data types and users are inside the database, and what external systems or limits bound it.

(b) Four limitations of mobile databases, with examples. [8 marks]

- Limited bandwidth and intermittent connectivity — a field worker may lose the network while updating a delivery record.
- Limited storage, memory and battery — a smartphone may not hold a full local copy of a large central database.
- Security and privacy risks — a lost or stolen mobile device may expose cached customer data unless it is encrypted.
- Synchronisation and consistency — offline edits from several devices must be merged without overwriting newer data.
- Heterogeneous devices and operating systems increase development and testing effort.

(c) Rows that will fail insertion into the STAFF table. [4 marks]

- Row 3033 (Siti Ziana) will fail because Salary = 11000 is outside the allowed range 3000–10000; this violates the domain or check constraint on Salary.
- Row 3034 (Ahmad Daniel) will fail because Salary = “150a” is not numeric; this violates the data type of the Salary column.
- Rows 3035 and 3036 satisfy the stated data types and range, and all StaffID values are unique.

Marking guide

- a) 2 marks for each of four initial-study activities explained (4 × 2 = 8).
- b) 2 marks for each of four mobile-database limitations with a relevant example (4 × 2 = 8).
- c) 2 marks for row 3033 with the range/domain reason; 2 marks for row 3034 with the data-type reason (2 + 2 = 4).

Provenance: model_unofficial · confidence high · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.

Knowledge base basis: TOPIC 2 DATA MODEL (1).pdf; TOPIC 6 DATABASE DESIGN (1).pdf

Assumptions / notes

- The staff table has StaffID as its primary key; Salary must be numeric and lie in the stated range 3000–10000; DepartmentID is numeric.

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QUESTION 3 (20 marks)

Source: ICT450-2026-FEB · page [3] · bank id ICT450-2026-FEB-Q3

Reference answer

Table 1 repeats the registration and volunteer data for each skill, so the skill list is a repeating group.

(a) Two problems in Table 1. [3 marks]

- RegisterID, RegisterDate, VolunteerID and VolunteerName repeat for every skill, creating redundancy.
- Skill and YearsOfExperience are stored in a repeating group rather than in a separate registration-skill relation.
- The blank VolunteerName cells also show inconsistent data entry.

(b) Keys and dependencies. [6 marks]

Flattened relation.

```
REGISTRATION_SKILL (RegisterID, RegisterDate, VolunteerID, VolunteerName, SkillID, SkillName,
YearsOfExperience)
```

Functional dependencies.

```
RegisterID → RegisterDate, VolunteerID
VolunteerID → VolunteerName
SkillID → SkillName
RegisterID, SkillID → YearsOfExperience
```

(c) Normalise to 3NF. [11 marks]

1NF: one row per registration-skill pair; key (RegisterID, SkillID).

2NF: remove volunteer, registration and skill partial dependencies.

2NF/3NF relations.

```
VOLUNTEER (VolunteerID, VolunteerName)
REGISTRATION (RegisterID, RegisterDate, VolunteerID)
SKILL (SkillID, SkillName)
REGISTRATION_SKILL (RegisterID, SkillID, YearsOfExperience)
```

3NF: VolunteerName depends on VolunteerID and SkillName on SkillID; no further transitive dependency remains.

Sample 3NF data: REGISTRATION

RegisterID	RegisterDate	VolunteerID
101	28/9/2025	V15
102	5/10/2025	V17
103	30/10/2025	V20

Sample 3NF data: REGISTRATION_SKILL

RegisterID	SkillID	YearsOfExperience
101	J1	5
101	S2	2
102	M1	25
103	W1	10

Marking guide

- a) 3 marks: two problems with reasons.
- b) 6 marks: key and dependencies.
- c) 11 marks: 1NF (2), 2NF (4), 3NF (5).

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Knowledge base basis: TOPIC 5 NORMALIZATION (1).pdf

Assumptions / notes

- RegisterID identifies one registration; each registration can list several skills. Years of experience belongs to the registration-skill pair.

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QUESTION 4 (20 marks)

Source: ICT450-2024-JUL · page [5] · bank id ICT450-2024-JUL-Q4

Reference answer

Relational schema used by the question.

```
STUDENT (StudentID, FirstName, LastName, PhoneNumber, FacultyID*)
CANDIDATE (CandidateID, StudentID*)
VOTING (VotingID, VotingDate, StudID*)
VOTING_DETAILS (VotingID*, CandidateID*, VotingTime)
FACULTY (FacultyID, FacultyName)
```

(a) Total number of candidates for each faculty. [4 marks]

```
SELECT f.FacultyID, f.FacultyName,
       COUNT(c.CandidateID) AS total_candidates
FROM FACULTY f
JOIN STUDENT s   ON s.FacultyID = f.FacultyID
JOIN CANDIDATE c ON c.StudentID = s.StudentID
GROUP BY f.FacultyID, f.FacultyName;
```

(b) Students who do not vote for any candidate. [4 marks]

```
SELECT s.*
FROM STUDENT s
WHERE NOT EXISTS (
    SELECT 1
    FROM VOTING v
    JOIN VOTING_DETAILS vd ON vd.VotingID = v.VotingID
    WHERE v.StudID = s.StudentID
);
```

(c) Total number of voters for each faculty. [4 marks]

```
SELECT f.FacultyID, f.FacultyName,
       COUNT(DISTINCT v.StudID) AS total_voters
FROM FACULTY f
JOIN STUDENT s ON s.FacultyID = f.FacultyID
JOIN VOTING v  ON v.StudID = s.StudentID
GROUP BY f.FacultyID, f.FacultyName;
```

(d) Full name of candidates with more than 500 votes. [4 marks]

```
SELECT c.CandidateID,
       CONCAT(s.FirstName, ' ', s.LastName) AS full_name,
       COUNT(vd.VotingID) AS total_votes
FROM CANDIDATE c
JOIN STUDENT s       ON s.StudentID = c.StudentID
JOIN VOTING_DETAILS vd ON vd.CandidateID = c.CandidateID
GROUP BY c.CandidateID, s.FirstName, s.LastName
HAVING COUNT(vd.VotingID) > 500;
```

In Oracle, replace CONCAT(...) with s.FirstName || ' ' || s.LastName.

(e) Total number of students who vote for only one candidate. [4 marks]

```
SELECT COUNT(*) AS students_voting_once
FROM (
    SELECT v.StudID
    FROM VOTING v
    JOIN VOTING_DETAILS vd ON vd.VotingID = v.VotingID
    GROUP BY v.StudID
    HAVING COUNT(DISTINCT vd.CandidateID) = 1
) AS single_vote;
```

Marking guide

- a) 4 marks: candidate-student-faculty join (2), COUNT per faculty (2).
- b) 4 marks: anti-join/not-exists on voting details (3), correct student output (1).
- c) 4 marks: faculty-student-voting join (2), COUNT DISTINCT voters (2).
- d) 4 marks: candidate vote join (2), GROUP BY and HAVING > 500 (2).
- e) 4 marks: group votes by student (2), HAVING count = 1 and outer COUNT (2).

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Knowledge base basis: TOPIC 7 SQL.pdf

Assumptions / notes

- A candidate's faculty is the faculty of the student who is the candidate.
- A vote is represented by a VOTING row plus its VOTING_DETAILS rows; a student who has no VOTING_DETAILS row has not voted for a candidate.

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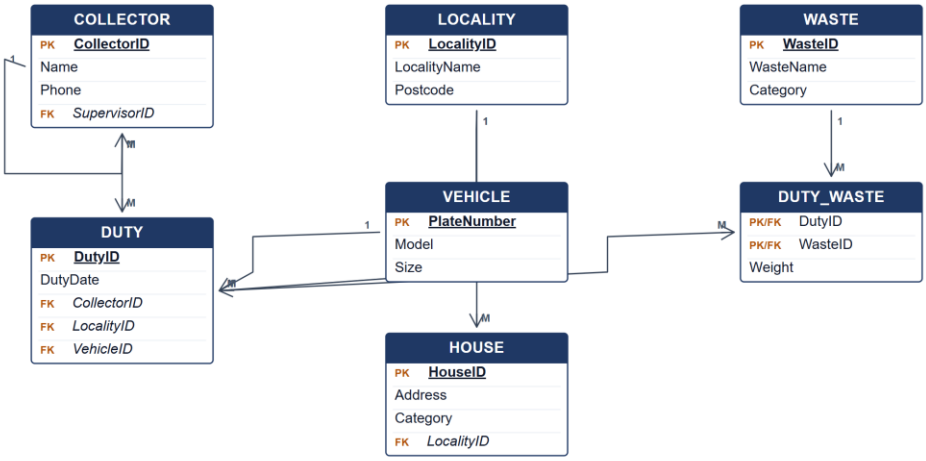
QUESTION 5 (20 marks)

Source: ICT450-2025-JUL · page [5, 6] · bank id ICT450-2025-JUL-Q5

Reference answer

The ERD models collectors, duties, localities, vehicles, houses and waste types. The supervisor relationship is recursive on COLLECTOR, and DUTY_WASTE records the weight collected for each waste type.

(a) Construct the ERD with Crow's Foot notation. [16 marks]



Clean Environment ERD.

(b) Two pieces of information from the ERD. [4 marks]

- The collection duties assigned to each collector, with date, locality and vehicle.
- The total weight collected for each waste type and locality.

Marking guide

- a) 16 marks: collector/duty/locality/vehicle (6); house and waste (4); recursive supervisor and DUTY_WASTE relationships (4); keys (2).
- b) 2 marks for each information item (4).

Provenance: model_unofficial · confidence medium · Model-authored from the approved question text, the supplied source assets and the ICT200/ICT450 knowledge base; no official ICT450 marking scheme was supplied.
Knowledge base basis: TOPIC 4 ERD_stu (2).pdf; TOPIC 3 RELATIONAL DATA MODELstu.pdf

Assumptions / notes

- Supervisor is a recursive role on COLLECTOR; a duty collects many waste types with a weight per waste type.

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